

Gata4 and *Gata5* Cooperatively Regulate Cardiac Myocyte Proliferation in Mice*

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GATA5 is a member of the zinc finger transcription factor GATA family (GATA1–6) that plays a wide variety of roles in embryonic and adult development. Experiments in multiple model systems have emphasized the importance of the GATA family members 4–6 in the development of the endoderm and mesoderm. Yet despite overlapping expression patterns, there is little evidence of an important role for GATA5 in mammalian cardiac development. We have generated a new *Gata5* mutant allele lacking exons 2 and 3 that encodes both zinc finger domains (*Gata5*^{tm2Em}), and we show that although *Gata5*^{−/−} mice are viable, *Gata4*^{+/−}*Gata5*^{−/−} mutants die at mid-gestation and exhibit profound cardiovascular defects, including abnormalities of cardiomyocyte proliferation and cardiac chamber maturation. These results demonstrate functional redundancy between *Gata4* and *Gata5* during cardiac development and implicate *Gata5* as a candidate modifier gene for congenital heart disease.

All six members of the GATA family (GATA1–6) of zinc finger transcription factors play important roles in cell fate decision, differentiation, and morphogenesis (1). Members of this family recognize the GATA motif, which is present in the promoters of many genes. These factors have been divided into two subfamilies, *Gata1/2/3* and *Gata4/5/6*, based on their expression patterns and amino acid sequence homology (2). *Gata1*, *Gata2*, and *Gata3* are preferentially expressed in hematopoietic cells and regulate proliferation and differentiation during hematopoiesis (3). *Gata4*, *Gata5*, and *Gata6* are predominantly expressed during embryonic heart development, in addition to other sites. The DNA binding domains of GATA4, GATA5, and GATA6 are ~85% similar at the protein level. During early cardiac development, *Gata4* expression is confined to the cardiac crescent,

and later the transcripts are detected throughout the myocardium and endocardium (4, 5). *Gata4*-deficient mice die between E8.5⁴ and E10.5 because of defects in ventral morphogenesis, including a failure of the cardiac mesoderm to form a linear heart tube (6, 7). A frameshift mutation in human *GATA4* (E359del) is linked to cardiac septal defects (8). Like *Gata4*, *Gata6* is expressed in the precardiac mesoderm and later in myocardial cells. *Gata6* is also expressed in vascular smooth muscle (9). *Gata6* null mice die prior to cardiac development because of defects in the visceral endoderm function and extraembryonic development (10, 11).

The temporal and spatial expression pattern of *Gata5* suggests its involvement in tissue-specific transcriptional regulation during embryonic development (12). *Gata5* is expressed in the precardiac crescent between E7 and E8 (12). By E9.5 in the heart, *Gata5* is expressed at high levels in the atria with lower levels observed in the ventricle and outflow tract. By E12.5, expression is primarily restricted to endocardial cells of the atria, and by E16.5, *Gata5* transcripts are no longer detected in cardiac tissues (12). In chick, *GATA5* is transcribed in the cardiac crescent prior to formation of the primordial heart tube (13). In *Xenopus*, *gata5* is expressed in both cardiac mesoderm and hepatogenic endoderm. Down-regulation of *gata5* using two nonoverlapping translation-blocking and splice site-blocking morpholino causes severe reduction in the number of heart and liver precursors at the time of or shortly after their specification (14). In zebrafish, *gata5* is encoded by the *faust* locus, and mutants reveal that *gata5* is required for the production of normal numbers of developing myocardial precursors (15, 16). In addition, *gata5* transcriptionally regulates several cardiac genes important in cardiac development, including *nkx2.5* (14).

In a previous study, Molkenkin *et al.* (17) reported that disruption of the mouse *Gata5* gene had no effect on cardiac development or morphology. Homozygous *Gata5* mutants (*Gata5*^{tm1Eno}) are viable and fertile, with female mice harboring a genitourinary abnormality. Considering the prominent and essential role that *Gata5* appeared to play in the development of the heart in other model systems, this was a surprising result (14, 18). Indeed, subsequent analysis of potential *Gata5* isoforms expressed in disparate model systems suggests that the *Gata5*^{tm1Eno} allele might be hypomorphic. For example, the

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⁴ The abbreviations used are: E, embryonic day; ChIP, chromatin immunoprecipitation; TUNEL, terminal dUTP nick-end labeling; WT, wild type; RT-qPCR, reverse transcription-quantitative PCR; ES, embryonic stem.

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chick harbors a novel *GATA5* isoform that contains a DNA binding domain composed of a single zinc finger that can bind to DNA and promote transcription (19). This is a common feature in other GATA family members with mouse *Gata1*, human and mouse *Gata2*, human and mouse *Gata3*, and human and mouse *Gata6* genes all possessing two promoters and two initiation codons (20–24). Analysis of the previously targeted *Gata5*^{tm1Eno} allele suggests that a partially functional transcript analogous to this isoform could be produced. Thus, we sought to generate a novel *Gata5* allele to assess the complete loss of *Gata5* in mammals.

Here, we report the creation and characterization of a new *Gata5* mutant allele lacking both zinc finger domains (*Gata5*^{tm2Eem}, hereafter designated as *Gata5*^{−/−}), and we demonstrate striking functional redundancy between *Gata4* and *Gata5* during cardiac development. Murine embryos are exquisitely sensitive to the loss of a single *Gata5* allele in the setting of *Gata4* heterozygosity resulting in profound cardiac defects, in part, due to the cooperative regulation of cardiomyocyte proliferation.

EXPERIMENTAL PROCEDURES

Targeting of the Mouse *Gata5* Locus and Generation of *Gata4/5* Compound Mutants—A *Gata5* targeting vector was constructed in pPNT containing pGK-Neo and pGK-TK cassettes for positive and negative selection, respectively (25). The 5' arm was a 1.9-kb fragment of *Gata5* genomic DNA that includes the majority of exon1, and 20 bp of exon2 (first zinc finger) was cloned upstream of *lacZ*. The 3' arm was an 8-kb genomic fragment beginning immediately after exon 3 (second zinc finger) (Fig. 1A). The targeting vector was linearized with *SacII* and electroporated into R1 ES cells (26). After 24 h, neomycin-resistant colonies were selected in 250 μg/ml G418 and 1.5 M ganciclovir for 7 days. Genomic DNA from resistant ES cell clones was isolated and analyzed by Southern blot after *KpnI* digestion (Fig. 1B). Hybridization of the probe to *KpnI*-digested DNA yielded a 29.8-kb wild-type band and a 6.2-kb mutant band. Seven of 200 individual ES cell colonies contained the 6.2-kb mutant band for a targeting efficiency of 3.5%. Positive clones were microinjected into E3.5 mouse C57BL/6 blastocysts and subsequently implanted into pseudo-pregnant CD1 females using standard techniques. Male chimeras were mated with C57BL/6 females to obtain agouti germ line *Gata5* heterozygotes. *Gata5* heterozygotes were mated to generate *Gata5* mutant mice (*Gata5*^{tm2Eem}). Male *Gata5*^{−/−} were mated with female *Gata4*^{+/-} mice (7) to generate *Gata4*^{+/-}*Gata5*^{−/−} compound mutants. Further mating between male *Gata5*^{−/−} with female *Gata4*^{+/-}*Gata5*^{−/−} compound mutants generated *Gata4*^{+/-}*Gata5*^{−/−} mutant mice.

PCR Genotyping—Agouti offspring from *Gata5*^{+/-} intercrosses were genotyped by Southern blot or PCR analysis (Fig. 1, B and C). Genomic DNA was analyzed by Southern blot as described above. The following primers (listed 5' to 3') were used for PCR genotyping analysis: primer a, GGCATCTACCTTAGAGTGTGG; primer b, GCTCCCATCTTTAATCATCCT; and primer c, TAGGTTACGTTGGTGTAGATGGG. Primers a and b flank exon2 (first zinc finger) in introns 1 and 2, respectively, and amplify a 500-bp fragment from the wild-type *Gata5* gene. Primer c is a reverse primer in *lacZ* and

together with primer a amplifies a 750-bp fragment from mutant *Gata5* mice. *Gata4*^{+/-}*Gata5*^{−/−} mutants from *Gata5*^{−/−} crosses with *Gata4*^{+/-}*Gata5*^{+/-} were genotyped by PCR and Southern blot, as described previously and above (10).

Histology and in Situ Hybridization—Histology was performed as described previously (27, 28). Briefly, E12.5 and E14.5 mouse embryos were harvested from timed matings, fixed in 4% paraformaldehyde at 4 °C overnight, dehydrated through graded ethanol (30, 50, 70, 95, and 100%), embedded in paraffin, sectioned at 8 μm, and stained with hematoxylin and eosin according to standard protocol (12). Radioactive *in situ* hybridization was performed on paraformaldehyde-fixed, paraffin-embedded heart sections according to standard protocols (29). A 601-nucleotide *in situ* probe was generated containing the 3'-untranslated region of the mouse *Gata5* gene. The following primers (listed 5' to 3') were used to generate this fragment: forward, AGCCCCCAGGCTGGTCTCAG; reverse, CAGAAGTCTGCGATGATGGG.

Quantitative Real Time PCR and Microarray—RT-qPCR was performed as described previously (30). Briefly, total RNA was isolated from dissected mouse hearts at E12.5 and E14.5 using TRIzol (Invitrogen). RNA was reverse-transcribed using random hexamers and the SuperScript First Strand Synthesis kit (Invitrogen). Gene expression was measured by quantitative RT-PCR (ABI PRISM 7900) using SYBR Green (Applied Biosystems) or ABI Taqman chemistry (*Gata4*, Mm03053570_s1; *Gata5*, Mm03053574_s1; and *Gata6*, Mm00802636_m1). Signals were normalized to corresponding glyceraldehyde-3-phosphate dehydrogenase controls, and the ratios expressed as fold changes when compared with wild-type controls. PCR conditions and primer set sequences are available upon request. Microarray analysis was performed on RNA isolated from E10.5 *Gata5*^{+/-} (*n* = 3), *Gata5*^{−/−} (*n* = 3), and *Gata4*^{+/-}*Gata5*^{−/−} (*n* = 3) hearts using Affymetrix GeneChip mouse 430 2.0 microarray. Differences in gene expression levels were considered significant at *p* < 0.05, and raw data were deposited in GEO.

ChIP Assay—Chromatin was isolated from E12.5 murine hearts using a commercially available kit (Upstate Biotechnology, catalogue no. 17-295). In brief, the entire heart was excised from the embryo (including systemic and pulmonary venous attachments, atria, ventricles, and proximal great vessels), manually minced, and fixed in 1% formaldehyde. Chromatin was sheared by sonication to an average length of 200–500 bp, immunoprecipitated with either a GATA5- or GATA4-specific antibody (Santa Cruz Biotechnology, M-20, C-20, respectively) or with an IgG negative control. Reverse, cross-linked immunoprecipitated chromatin was then subjected to PCR analysis using primers listed as follows (*gene*/GATA-binding site position with respect to transcription start site/product size (bp))/forward PCR primer sequence/reverse PCR primer sequence: *Cdkn1a*/−195/93, ACACTTCCTCTTCCTTCCTGGGTAGGGCTGATAGAAAGGAAATCCAAGGG; *Cdkn1b*/−1158/133, ACGACAGGCCTGGCTATTCTGTTTACCACCACCACGGTATTACCAACA; *Cdkn1c*/−636/80, CTCTACACCCAGGCCTAAGTATCTGTCTAACTCCGTGATAGAGGGTCTTAGGG; and *Cdkn2c*/−1460, −1431/98, GGAAGCTATGATAGGAGATTGGCACCCAAGCAAGACACCA-TTAATTTCCAG.

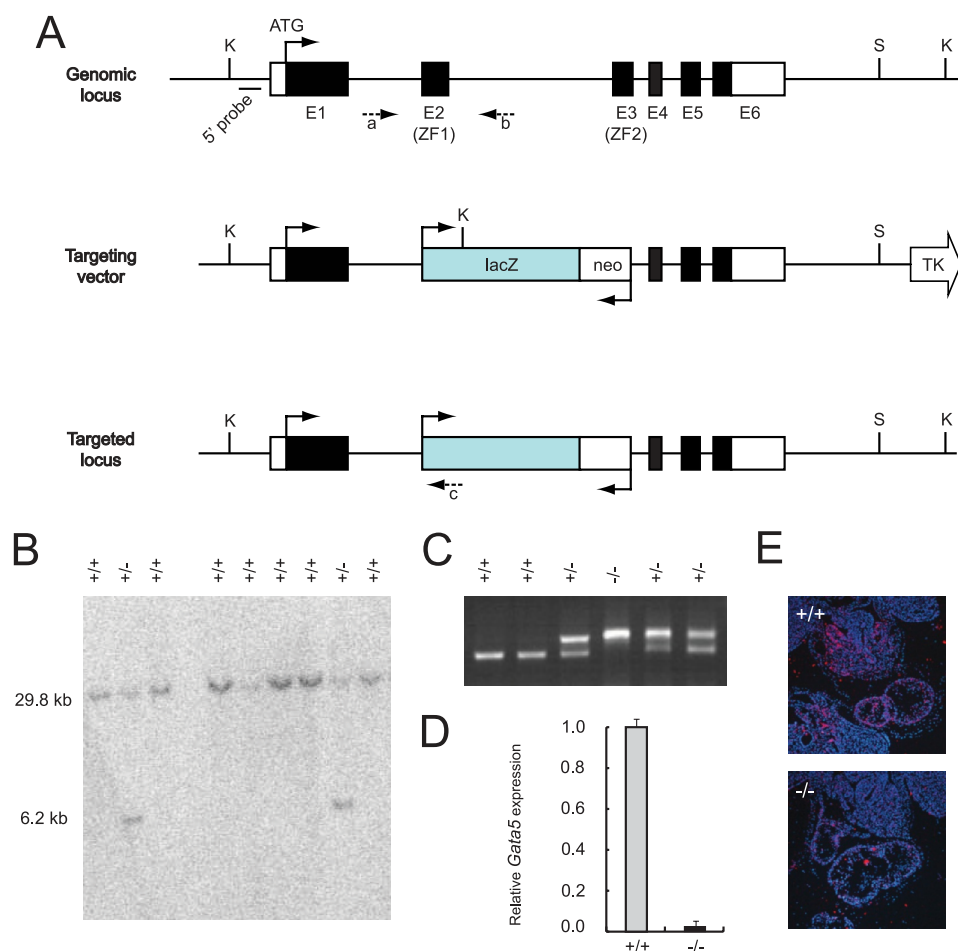


FIGURE 1. Generation and confirmation of *Gata5* mutant mice. *A*, *Gata5* null mice were generated by standard gene-targeting techniques using homologous recombination in murine R1 ES cells. Exons 2 and 3 were replaced with a LacZ-pGK-Neo cassette. *K*, KpnI; *S*, SalI; *TK*, thymidine kinase; *c*, primer *c* as described under "Experimental Procedures." *B*, Southern blot with a 5' probe identified multiple positive ES clones that were used to generate chimeras. These were backcrossed with C57/Bl6 mice to identify germ line transmission. *C*, intercrossing *Gata5* heterozygous germ line mice gave rise to viable and fertile *Gata5* nulls identified by PCR with wild-type primers "a and b" and mutant primers "a-c". *D*, RT-qPCR for *Gata5* transcripts revealed no accumulation indicating that the new allele was a complete null. *E*, null allele was further confirmed by *in situ* hybridization at E10.5 revealing a complete absence of *Gata5* transcript in all tissues.

Proliferation and TUNEL Analyses—Cardiomyocyte proliferation was evaluated by immunohistochemistry on WT, *Gata5* mutant, and *Gata4/5* compound mutant heart sections by Ki67 (1:50), phosphohistone H3 (1:2000, Cell Signaling, Beverly, MA), and MF20 (1:25, Developmental Studies Hybridoma Bank, University of Iowa, Iowa City). DAPI (Vector Laboratories, Burlingame, VT) was used to stain nuclei. ImageJ software was used to count Ki67 and phosphohistone H3-positive myocyte nuclei and the total number of myocytes in 6–8 different sections of 3–4 independent heart samples. Apoptosis was assayed by TUNEL assay using the manufacturer's instructions (Invitrogen).

Statistics—All measurement data are expressed as mean $\pm$ S.D. The statistical significance of differences between groups was analyzed by Student's *t* test. Differences were considered significant at a *p* value <0.05 .

RESULTS

***Gata5*^{tm2Em} Null Mice Are Viable, Fertile, and Demonstrate Normal Cardiac Morphology**—To elucidate the function of GATA5 in cardiac development, we generated *Gata5* null mice

by standard gene-targeting techniques using homologous recombination in murine ES cells (25). A new *Gata5* allele (*Gata5*^{tm2Em}) was created with a targeting construct that substituted both zinc finger domains (exons 2 and 3) with a pGK-Neo cassette (Fig. 1*A*). Southern blot and PCR analysis of 200 individual ES clones confirmed appropriate integration of the targeting construct in seven clones, representing a 3.5% targeting efficiency (Fig. 1, *B* and *C*). Quantitative RT-PCR and *in situ* hybridization analysis confirmed that the targeted modification of the *Gata5* locus resulted in a functionally null allele (Fig. 1, *D* and *E*). Mice heterozygous for the mutant *Gata5* allele appeared normal and fertile. Offspring from *Gata5*^{+/−} intercrosses resulted in an expected Mendelian ratio of 18 (26%) *Gata5*^{+/+}, 36 (52%) *Gata5*^{+/−}, and 15 (22%) *Gata5*^{−/−} (*p* = 0.822). *Gata5* null males were phenotypically normal, whereas females exhibited genitourinary abnormalities identical to those exhibited by the *Gata5*^{tm1Eno} allele (17). *Gata5*^{−/−} females showed a reduction in the distance between the vagina and anus identical to that seen in the *Gata5*^{tm1Eno} allele with reduced fertility. Histological analysis of WT and *Gata5* mutant (*n* = 6) hearts from embryonic and adult stages revealed no

obvious cardiac defects (Fig. 3 and data not shown). Echocardiographic analysis of three-month-old *Gata5* null mice revealed normal cardiac function when compared with WT littermates (data not shown).

Compound *Gata4*^{+/−}*Gata5*^{−/−} Mutant Mice Are Embryonic Lethal and Display Profound Cardiac Defects—*Gata4*, *Gata5*, and *Gata6* have partially overlapping expression patterns and are transcribed in both extra-embryonic and embryonic mesoderm early in development (9, 13, 31). Transcript abundance for *Gata4*, *Gata5*, and *Gata6* was analyzed by RT-qPCR in *Gata5* null mice and wild-type littermates at E12.5 and E14.5 (Fig. 2). We noted a greater than 2-fold increased expression of *Gata4* and *Gata6* in *Gata5* null hearts suggesting the possibility of functional compensation.

To test the hypothesis that *Gata4* might compensate for *Gata5* function in *Gata5* mutant hearts, we generated *Gata4/5* compound mutants. Mice heterozygous for *Gata4* were viable and demonstrated no obvious cardiovascular phenotypes, consistent with previous reports (6, 7). *Gata4/5* compound heterozygotes (*Gata4*^{+/−}*Gata5*^{+/−}) were also viable and fertile. At

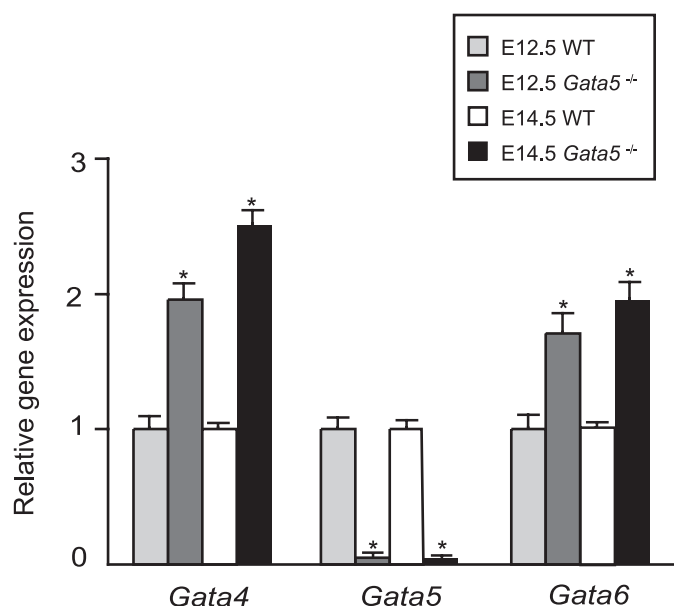


FIGURE 2. RT-qPCR of *Gata4*, *Gata5*, and *Gata6* in *Gata5* null embryos. *Gata4* and *Gata6* expression are significantly unregulated in whole heart extracts of *Gata5* null mice at two time points in midgestation, E12.5 and E14.5 ($n = 3$ per time point). *, $p < 0.05$.

E12.5, gross morphology of *Gata4*^{+/-}*Gata5*^{+/-} hearts is indistinguishable compared with WT controls. However, histological analysis of *Gata4*^{+/-}*Gata5*^{+/-} hearts demonstrated a 43% reduction in ventricular compact zone thickness as early as E12.5 (Fig. 3). By E14.5, differences in myocardial thickness between WT and *Gata4*^{+/-}*Gata5*^{+/-} hearts were also prominent with a 32% reduction in compact zone thickness (Fig. 3).

To determine whether complete loss of GATA5 increases the severity of cardiac phenotype, we generated *Gata4*^{+/-}*Gata5*^{-/-} compound mutant mice. No *Gata4*^{+/-}*Gata5*^{-/-} progeny was identified in newborn litters, suggesting that absence of a single *Gata4* allele in the presence of *Gata5* null is critical for embryonic development (Table 1). At E12.5, *Gata4*^{+/-}*Gata5*^{-/-} mutant embryos were present at a Mendelian ratio but harbored significant cardiac defects (Figs. 3 and 4). First, the ventricular, compact myocardium was hypoplastic (E12.5, 61% reduction in thickness with respect to WT; E14.5, 84% reduction in thickness with respect to WT) with a fine, abnormal trabecular structure (Fig. 3). Second, there was hypoplasia of the endocardial cushions by E12.5 leading to complete common atrioventricular canal (Fig. 4). This was occasionally observed as an unbalanced canal leading to a functionally univentricular heart (Fig. 4f). This abnormality was associated with a primum atrial septal defect as well as a large inlet-type ventricular septal defect modeling complete common atrioventricular canal. Third, the conotruncal cushions harbored defects in both decreased size and location leading to multiple abnormalities of ventriculoarterial positioning (Fig. 4, A–F). By E14.5, most of the *Gata4*^{+/-}*Gata5*^{-/-} embryos exhibited systemic hemorrhage and body wall edema suggestive of heart failure (data not shown).

Myocardial Proliferation and Apoptosis in Compound *Gata4*^{+/-}*Gata5*^{-/-} Mutant Hearts—GATA transcription factors have been previously implicated in regulating proliferation and

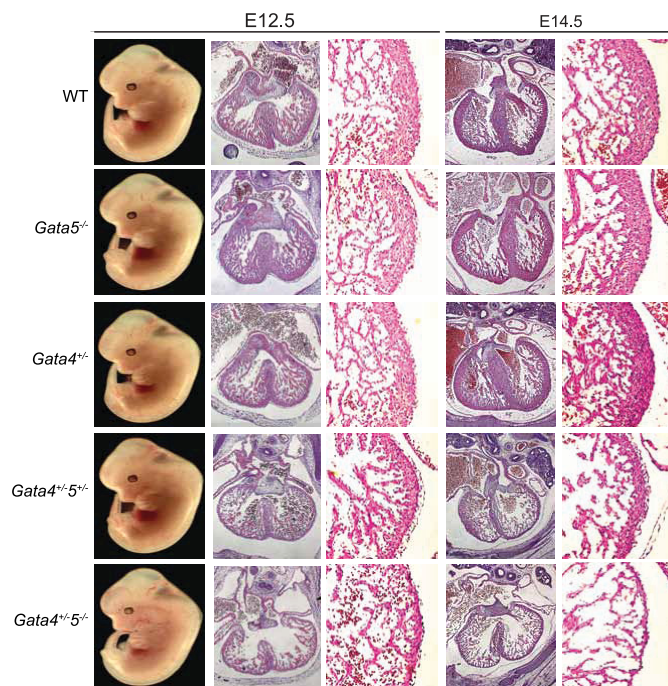


FIGURE 3. Ventricular phenotype of *Gata4/5* compound mutant mice. Compared with WT embryos, *Gata5*^{-/-} and *Gata4*^{+/-} (1st to 3rd rows, respectively) mice appeared grossly normal, whereas *Gata4*^{+/-}*Gata5*^{-/-} (5th row) compound mutants all died mid-gestation with no survivors past E14.5. There was progressive ventricular wall thinning, and the compact myocardium was hypoplastic with increasingly fine, abnormal trabecular structures with loss of each additional *Gata5* allele in the setting of *Gata4* heterozygosity (rows 3rd to 5th rows).

apoptosis. GATA4 has been shown to regulate myocardial proliferation and survival, whereas GATA6 inhibits cell cycle progression in vascular smooth muscle cells (32, 33). Ki67 was used to assess cell proliferation rates in WT and *Gata4*^{+/-}*Gata5*^{-/-} mutant hearts. *Gata4*^{+/-}*Gata5*^{-/-} mutant hearts showed 51% (at E12.5) and 55% (at E14.5) reduction in Ki67 staining throughout the ventricular myocardium when compared with WT controls (Fig. 5, A–E). There was also a trend toward decreased proliferation in the endocardial and epicardial compartments, although this observation did not reach significance to $p < 0.05$. Similar results were observed when proliferation was analyzed by phosphohistone H3 (Fig. 5F), yet there was no increase in apoptosis as evaluated by TUNEL staining (Fig. 5G). By contrast, proliferation in skeletal muscle was unchanged in the *Gata4*^{+/-}*Gata5*^{-/-} mutant embryos (data not shown). These experiments suggest that the two alleles of *Gata4* and *Gata5* are crucial for cardiomyocyte proliferation during embryonic development.

Altered Expression of Cell Cycle Genes in Compound *Gata4*^{+/-}*Gata5*^{-/-} Mutant Hearts—Cardiomyocytes proliferate during embryogenesis but have a severely decreased capacity soon after birth. Cell cycle progression is tightly controlled by both positive (*i.e.* cyclins and cyclin-dependent kinases) and negative regulators (*i.e.* Cdks-cyclin-dependent kinase inhibitors) (33–35). Recent experiments demonstrated that GATA4 is required for cyclin D2, *Cdk4*, and cyclin A2 expression in the ventricular myocardium. Furthermore, cyclin D2 and *Cdk4* are direct targets of GATA4 (32). To determine whether the ventricular myocardial hypoplasia observed in *Gata4*^{+/-}*Gata5*^{-/-}

TABLE 1

Viability of Gata4/5 compound mutant mice at four developmental time points

Numbers of embryos under Gata genotype portion of table refer to observed (and expected) pups.

Age	Total no.	Gata genotype				p value
		4 ^{+/+} 5 ^{+/-}	4 ^{+/-} 5 ^{+/-}	4 ^{+/+} 5 ^{-/-}	4 ^{+/-} 5 ^{-/-}	
>8 weeks	60	23 (15)	26 (15)	11 (15)	0 (15)	$p < 0.0001$
E14.5	16	4 (4)	4 (4)	3 (4)	5 (4)	$p = 0.97$
E12.5	91	24 (23)	21 (23)	24 (23)	22 (23)	$p = 0.26$
E10.5	79	18 (20)	27 (20)	16 (20)	18 (20)	$p = 0.30$

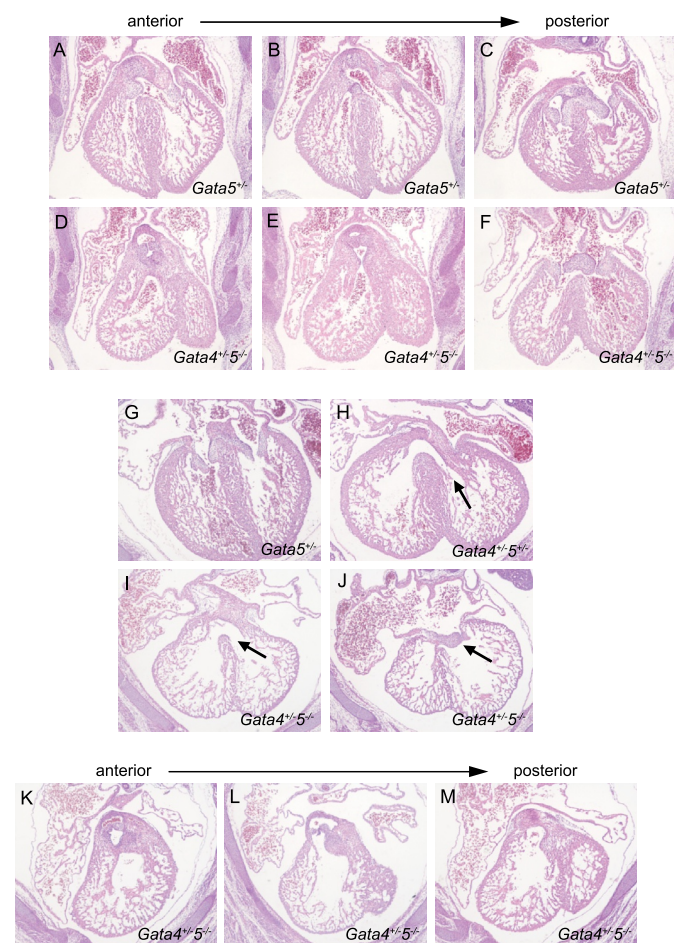


FIGURE 4. Conotruncal and endocardial cushion phenotype of E14.5. *Gata4/5* compound mutant mice. Compound *Gata4/5* mutant mice harbored multiple defects in distinct anatomic compartments. *Gata5*^{+/-} mice appeared normal with no identifiable intracardiac defects (A–C and G). *Gata4*^{+/-}5^{+/-} double heterozygotes were viable but demonstrated conotruncal septal defects with incomplete penetrance (H, arrow). *Gata4*^{+/-}5^{-/-} compound mutants demonstrated moderately hypoplastic and anteriorly malaligned left ventricular outflow tracts (D–F). Similarly, ventriculoarterial positioning of the right ventricular outflow tract was more posteriorly positioned than normal (K–M). There was hypoplasia of the endocardial cushions leading to complete common atrioventricular canal defects. This was associated with a primum atrial septal defect as well as a large ventricular septal defect modeling human complete common atrioventricular canal (I and J).

mutant hearts was due to altered expression of cell cycle regulators, we analyzed the abundance of cyclin D2, cyclin A2, *Cdk4*, *Cdkn2c* (p18^{INC4c}), *Cdkn1a* (p21^{WAF/CIP1}), *Cdkn1b* (p27^{Kip1}), and *Cdkn1c* (p57^{Kip2}) transcripts by quantitative RT-PCR on E12.5 and E14.5 heart samples from WT and *Gata4*^{+/-}5^{-/-} mutant embryos. Gene expression levels of cyclin D2, cyclin A2, and *Cdk4* were unchanged in *Gata4*^{+/-}5^{-/-} mutant hearts compare with WT controls (Fig. 6A). In contrast, expression

levels of the cell cycle inhibitors *Cdkn2c* (p18^{INC4c}), *Cdkn1a* (p21^{WAF/CIP1}), *Cdkn1b* (p27^{Kip1}), and *Cdkn1c* (p57^{Kip2}) were significantly increased in *Gata4*^{+/-}5^{-/-} mutant heart samples compared with WT controls (Fig. 6B). We next investigated if up-regulation of *Cdkn2c* (p18^{INC4c}), *Cdkn1a* (p21^{WAF/CIP1}), *Cdkn1b* (p27^{Kip1}), or *Cdkn1c* (p57^{Kip2}) in *Gata4*^{+/-}5^{-/-} mutants were reflected by direct binding by GATA4 or GATA5 in wild-type hearts. Bioinformatic analysis of 2 kb upstream of the transcription start site of four CDKI promoters revealed some potential conserved GATA factor-binding sites. We performed ChIP assay on WT E12.5 heart extracts and revealed that GATA4 and GATA5 both are recruited to cyclin-dependent kinase I promoters and thus likely directly regulate their gene expression *in vivo* (Fig. 7). Taken together, our quantitative expression analyses and ChIP results suggest that reduction in the myocardial cell proliferation of *Gata4*^{+/-}5^{-/-} mutant hearts is due to de-repression and early activation of cell cycle inhibitors.

DISCUSSION

The formation of vertebrate heart is a multistep process comprising patterning, cell differentiation, and morphogenesis (36). Transcription factors and their combinatorial activity have been shown to govern many of the underlying molecular pathways. GATA factors (GATA4, GATA5, and GATA6) are essential regulators of mesodermal and endodermal organ development (1). GATA transcription factors are among the first to be expressed in the developing heart and regulate different aspects of cardiac development. In rodents, GATA4 and GATA6 are critical for activation of the cardiomyocyte gene expression program (37). In contrast to GATA4 and GATA6, a role for GATA5 in vertebrate cardiomyocyte gene expression regulation and development has not been demonstrated. Using a new null allele of *Gata5* (*Gata5*^{tm2Eem}), we now show that *Gata5* acts cooperatively with *Gata4* to regulate cardiomyocyte proliferation and development. Our data demonstrate that in concert with *Gata4*, *Gata5* negatively regulates expression levels of the cell cycle inhibitors *Cdkn2c* (p18^{INC4c}), *Cdkn1a* (p21^{WAF/CIP1}), *Cdkn1b* (p27^{Kip1}), and *Cdkn1c* (p57^{Kip2}). In addition, we show that loss of a single allele of *Gata5* in *Gata4* heterozygotes leads to thinning of the ventricular myocardium, an intermediate cardiac phenotype compared with the profound defects seen in *Gata4*^{+/-}5^{-/-} mutant embryos. GATA5 now appears to act in concert with GATA4 consistent with their overlapping expression pattern during early cardiac development. Together, our data reveal a previously unappreciated role for GATA5 in cardiac proliferation and development.

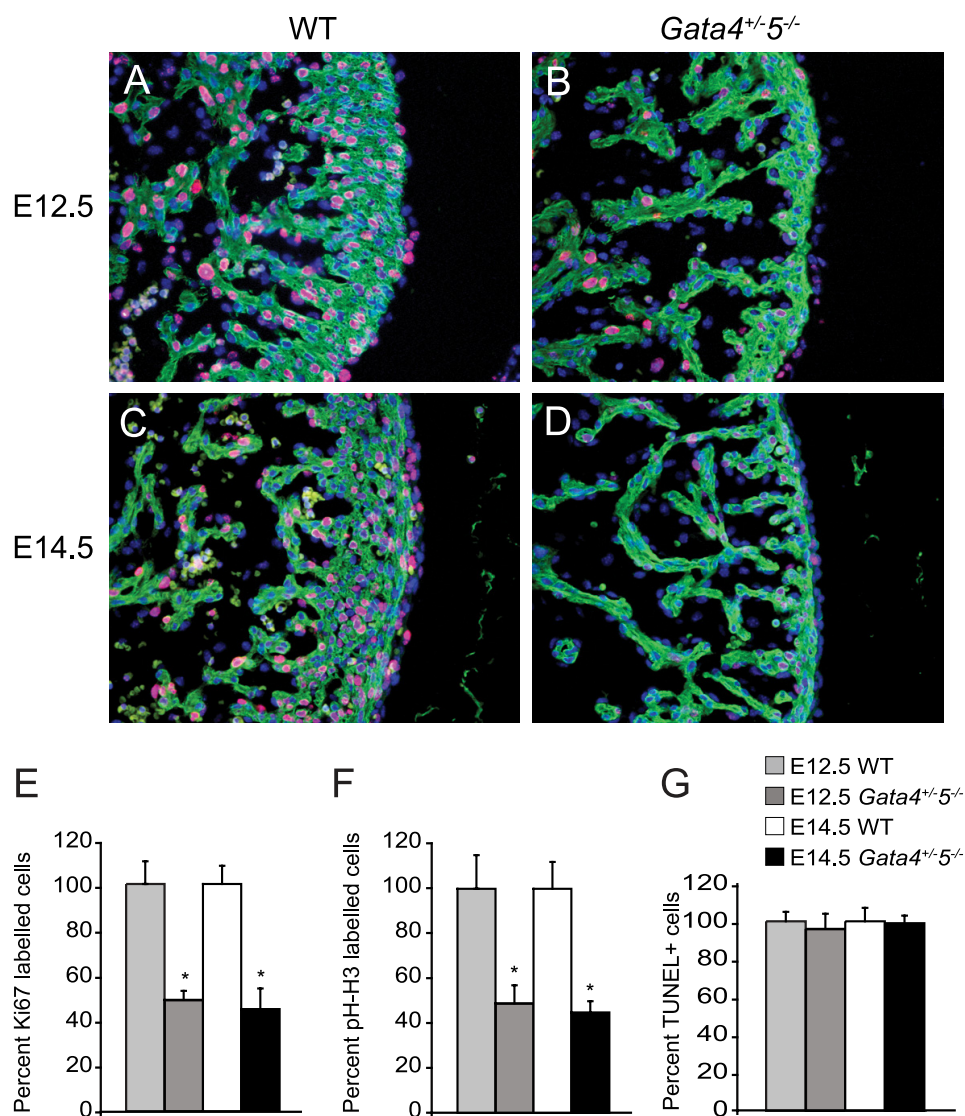


FIGURE 5. Cellular proliferation and apoptosis in *Gata4/5* compound mutant hearts. Cardiac myocytes in *Gata4*^{+/-}*5*^{-/-} compound mutant embryos (B and D) demonstrate a 51% (at E12.5) and 55% (at E14.5) reduced staining with either Ki67 or phosphohistone H3 (red) compared with WT embryos (A, C, and summarized in E), MF20 (green), and 4',6-diamidino-2-phenylindole (blue). Similar decreases in proliferation in *Gata4*^{+/-}*5*^{-/-} compound mutants results were observed by immunohistochemistry analysis with phosphohistone H3 (pH-H3, F). There was no change in apoptosis with respect to WT embryos in *Gata4*^{+/-}*5*^{-/-} compound mutants as assessed by TUNEL stain (G). *, *p* < 0.05.

GATA Factors and Cell Cycle Regulation during Mouse Cardiac Development—To ensure proper progression through the cell cycle, a series of orchestrated checkpoints modulate cyclin-dependent kinase (Cdk) complexes. Their activities are regulated in both a positive and negative manner (38). *In vivo* and *in vitro* data suggest that loss or down-regulation of CDK1 increases the proliferative capacity of neonatal cardiac myocytes (39, 40). GATA family of transcription factors has been previously implicated in cell cycle progression. GATA1 inhibits expression of *Cdk6* and cyclin D2 (*Ccn2*) and induces the cyclin-dependent kinase inhibitors *Cdkn2c* (p18^{INC4c}) and *Cdkn1b* (p27^{Kip1}) promoting proliferation arrest during erythroid maturation (34, 41, 42). GATA2 inhibits the growth of hematopoietic stem/progenitor cells by regulating expression levels of *Cdkn2a* (p21^{WAF1}) and *Cdkn1b* (p27^{Kip1}) (35, 43). Overexpression of *Gata6* inhibits cell cycle progression of vascular smooth

muscle cells by regulating expression of *Cdkn2a* p21^{WAF1} (33). Finally, GATA4 directly regulates expression levels of *Ccn2* (cyclin D) and *Cdk4* controlling myocardial cell proliferation in the anterior heart field (32).

In this study, we demonstrate that haploinsufficiency of both *Gata4* and *Gata5* result in cardiac defects characterized by thinning of the ventricular myocardial wall. In addition, loss of a second *Gata5* allele leads to profound cardiac defects including atrial septal defects, ventricular septal defects, ventricular hypoplasia, and malpositioning/hypoplasia of conotruncal and atrioventricular cushions associated with an increased expression of the cell cycle inhibitors *Cdkn2c* (p18^{INC4c}), *Cdkn1a* (p21^{WAF/CIP1}), *Cdkn1b* (p27^{Kip1}), and *Cdkn1c* (p57^{Kip2}). These data implicate GATA5 as a novel negative regulator of cyclin-dependent kinase I in the cardiac myocyte that helps balance the combined, important roles of GATA4–6 in cardiomyocyte proliferation and differentiation.

Functional Redundancy between GATA4 and GATA5 in the Heart—The transcription factors, GATA4–6, are ~89% identical in their zinc finger domain, recognize identical DNA sequence motifs, and are all expressed during early cardiac development (2). *Gata4* null mice that survive gastrulation form differentiated myocardium with increased *Gata6* expression suggesting compensation by *Gata6* for

loss of *Gata4* (6, 7). Although *Gata6* null mice undergo growth arrest during gastrulation due to an essential role of this factor in extra-embryonic endoderm development, ES cells deficient for *Gata6* are able to differentiate into cardiac myocytes *in vitro* and contribute to myocardium in chimeric embryos (37). As reported previously, *Gata5* null (*Gata5*^{tm1Eno}) did not exhibit any overt cardiac defects (17). In contrast to *Gata4* null mice, *Gata5*^{tm1Eno} null mice do not exhibit up-regulation of *Gata4* or *Gata6* expression. Despite intense scrutiny, we did not uncover any cardiac defects in our newly generated *Gata5* null (*Gata5*^{tm2Eem}) mice. However, we did identify increased expression of *Gata4* and *Gata6*, indicating a significant difference between the two alleles and suggesting that in the *Gata5*^{tm1Eem} mutant, *Gata4* and/or *Gata6* might compensate for the loss of *Gata5*. One possibility is that combined GATA4 and GATA5 function are predominantly mediated by the total

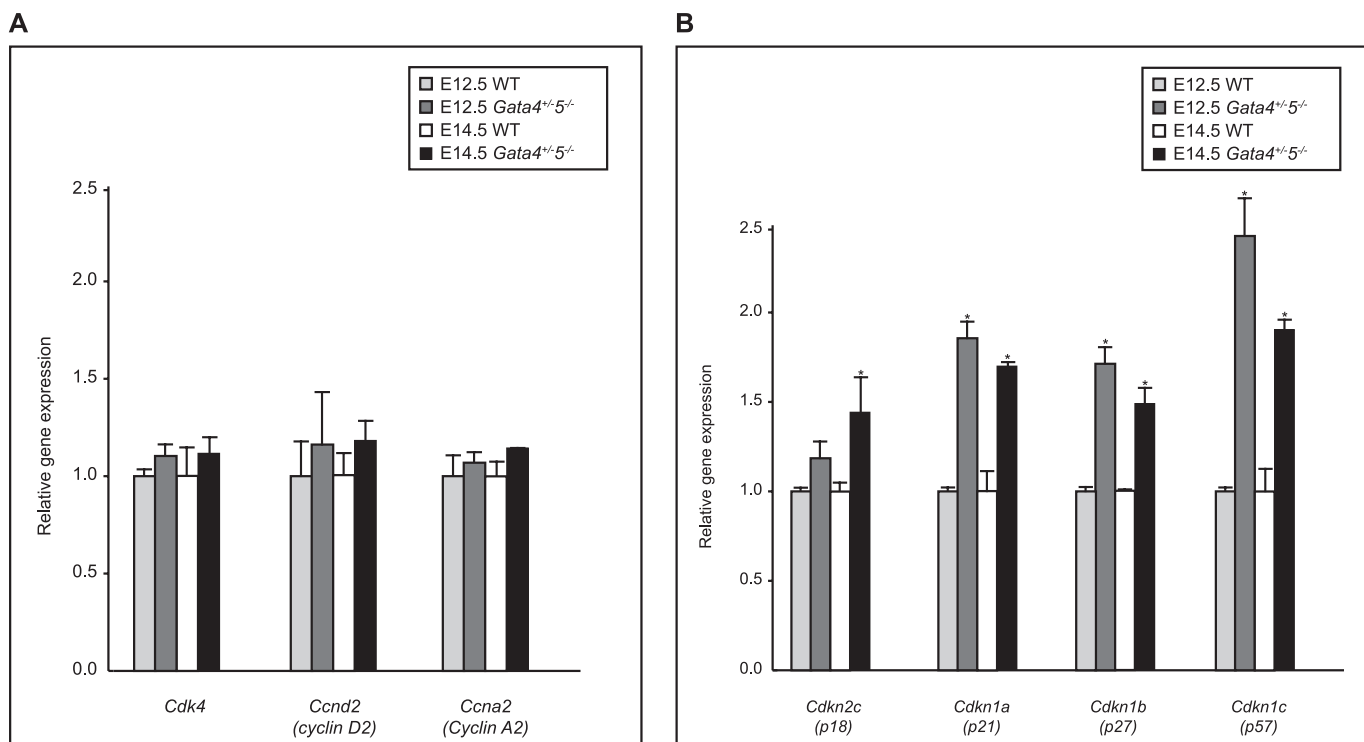


FIGURE 6. RT-qPCR of candidate cell cycle targets. A, RT-qPCR of candidate cyclin targets in cardiac extracts from E12.5 and E14.5 gene-targeted murine embryos. At two developmental time points, E12.5 and E14.5, *Cdk4*, *Ccnd2*, and *Ccna2* expression show no statistically significant differences to wild-type levels. B, at two developmental time points, E12.5 and E14.5, levels of *Cdkn1a*, *Cdkn1b*, and *Cdkn1c* transcripts are all increased in compound mutants relative to WT levels at E12.5. For *Cdkn2c*, RNA levels are increased at E14.5 (*, $p < 0.05$, $n = 3$).

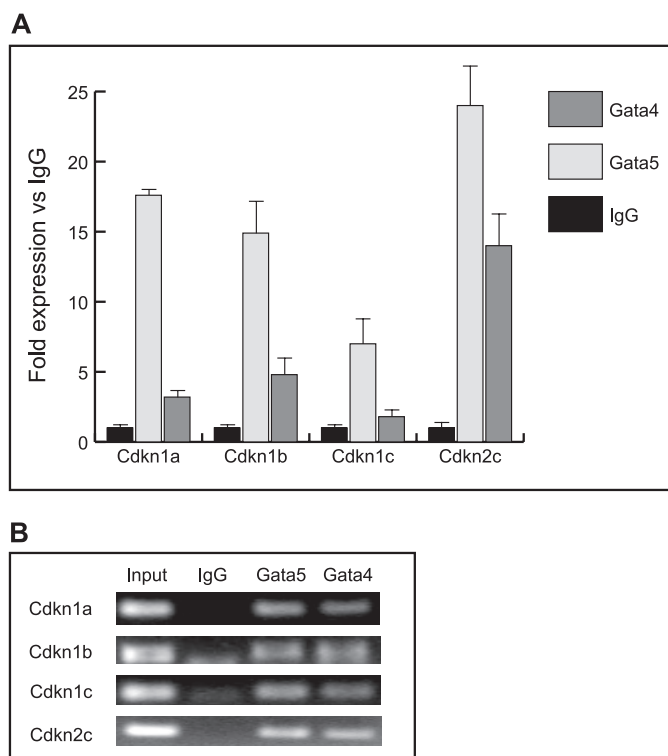


FIGURE 7. ChIP assay of GATA factor binding to cell cycle targets. ChIP assays demonstrate that GATA4 and GATA5 were associated with the mouse *Cdkn1a*, *Cdkn1b*, *Cdkn1c*, and *Cdkn2c* promoters. Chromatin DNA from "input" or individual transcription factor pulldown by IgG, GATA5, or GATA4 antibodies was amplified around the promoter region of *Cdkn1a*, *Cdkn1b*, *Cdkn1c*, and *Cdkn2c* containing the WGATAR motif(s). The littermate product is reported in fold IgG input (A) and by agarose gel electrophoresis (B).

level of GATA4, whereas in the absence of GATA4, GATA5 takes on a prominent role. Similarly, because GATA6 is also unregulated in *Gata5* null mice, albeit to a lesser degree than GATA4, there may be a combined compensatory up-regulation of both GATA4 and GATA6 messenger RNA. Another possibility is that GATA factor redundancy may be mediated by differential activity of the protein products. For example, Nemer and Nemer (44) suggest a specific interaction between GATA4 and GATA6 to regulate target genes, and a similar interaction might exist between GATA4/GATA5 or GATA5/GATA6.

Functional redundancy between GATA4 and GATA6 in the murine heart has been addressed by generating compound heterozygous *Gata4*^{+/-}*6*^{+/-} mice. Compound heterozygous *Gata4*^{+/-}*5*^{+/-} embryos die during embryonic development due to profound vascular and cardiac defects, including outflow tract defects, thin myocardium, and ventricular septal defects, whereas a complete loss of both *Gata4* and *Gata6* results in a loss of cardiac specification (37, 45). Thus, there is ample evidence for functional redundancy between *Gata4* and *Gata6* in mammalian cardiac development.

Despite these important findings, functional redundancy between GATA4 and GATA5 is less understood. In a previous study, made an effort to determine whether GATA4 or GATA6 substitute GATA5 functions by generating compound *Gata4*^{+/-}*5*^{+/-} and *Gata5*^{+/-}*6*^{+/-} mutants using the *Gata5*^{tm1Eno} allele (17, 45). Compound mutant hearts appeared to be identical to *Gata5* null (*Gata5*^{tm1Eno}) mice. This might have been due to residual expression of a truncated GATA5 protein (from an alternative promoter) in *Gata5*^{tm1Eno} null mice. Alternatively, there might be strain differences between

the analyses of the *Gata5*^{tm1Eno} and *Gata5*^{tm2Eem} alleles. In this study, we clearly demonstrate that compound heterozygous mice for *Gata4/5* (*Gata4*^{+/-}*5*^{+/-}) survive and exhibit thinning of the ventricular myocardium. However, further loss of the second allele of *Gata5* in compound mutant *Gata4*^{+/-}*5*^{-/-} mice leads to embryonic lethality. No viable *Gata4*^{+/-}*5*^{-/-} animals were ever observed after E14.5, indicating that this *Gata4/5* genotypic combination results in embryonic lethality with complete penetrance by mid-gestation. Thus, our data support the notion that *Gata5* plays an important role in vertebrate cardiac development, in part, by acting cooperatively with *Gata4*.

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